Q3_1

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I am implementing a two-sample t-test to determine whether people in countries with high GDP per capita (higher than the mean) are more happy or less happy than people in countries with low GDP (lower than the mean). I make the following assumptions for a t-test:

1. The data must be normally distributed
2. The data must be independent and identically-distributed
3. We are only comparing two groups (low and high gdp)

By inspecting Figure 1, we can see that the countries with a GDP lower than the mean tends to be less happy.

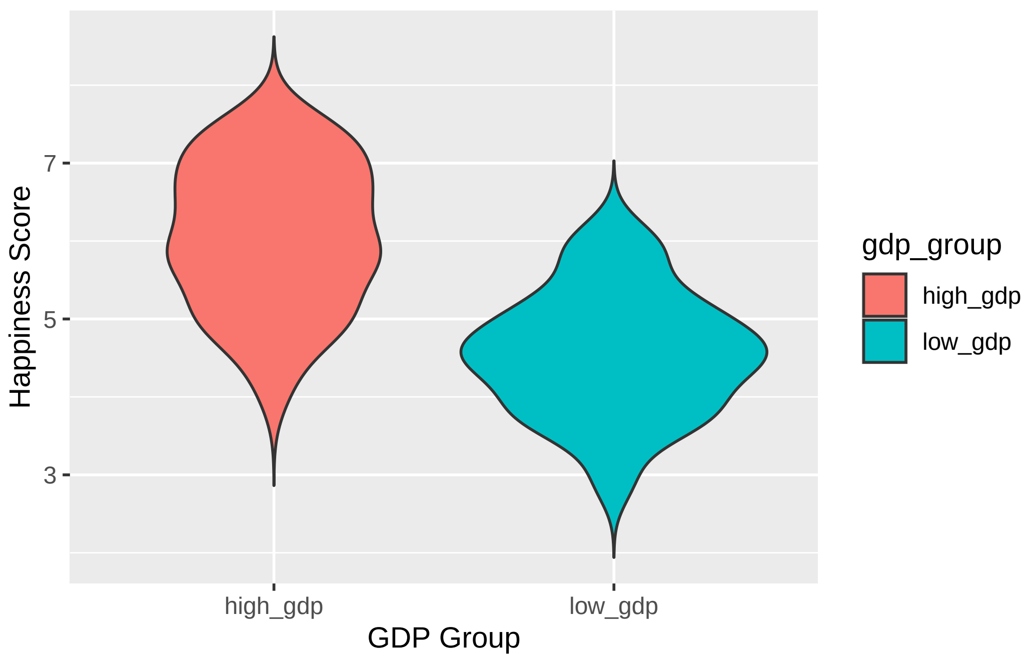

Figure 1: Distribution of Happiness Score by GDP Group

The result of the t-test with a T-statistic of 10.9851568 shows that there is a significant difference in happiness score for low and high GDP countries and a $p - value$ of $3.7519215 \times 10^{-21}$shows that the T-statistic is statistically significant.